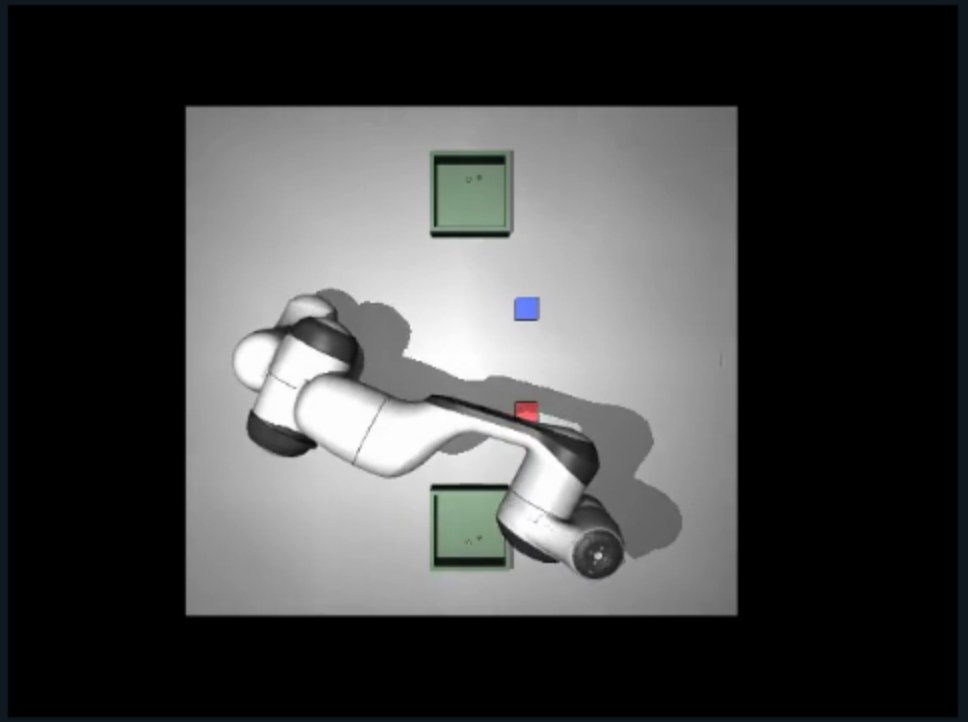
SimToReal Agent

Test before you touch.

Robot plan testing with AMD inference



Visionary Creators
Anish Jaiswal and Shailesh
AMD Developer Hackathon: ACT III

The operator’s problem

A valid JSON plan can still drop a part

Factory operators need evidence before accepting an AI-generated motion plan.

Our prototype tests each candidate in a Panda simulation and explains why it passes or fails.

The operator workflow

01 **Describe the task**

AMD Token Factory generates eight JSON candidates.

02 **Test independent scenes**

Local MuJoCo workers execute the plans and record evidence.

03 **Reject failures, then rank**

Measured checks cover task success, table collisions and drops.

04 **Review and approve**

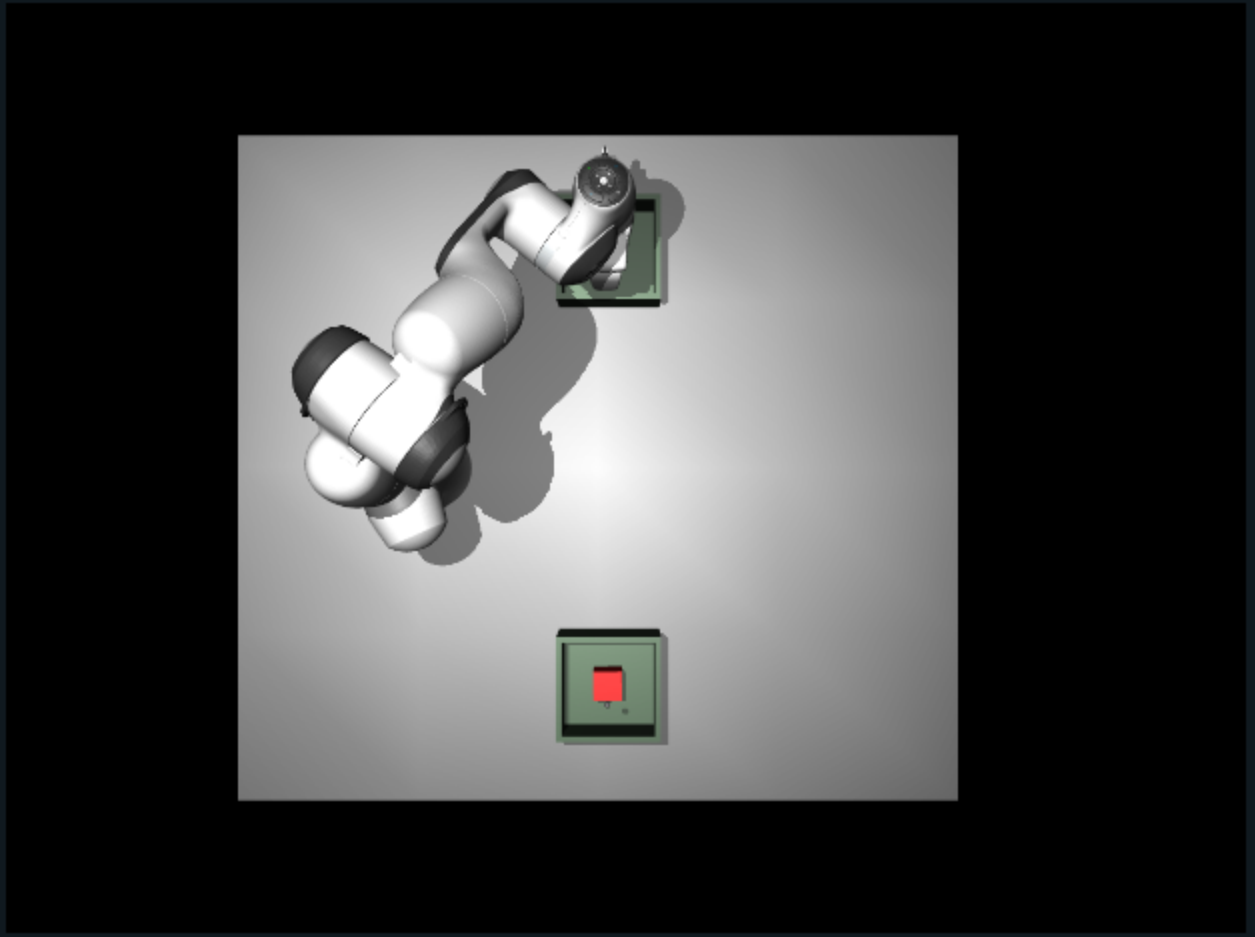
A camera check flags uncertainty. The operator approves export.

Demo: sorting parts and rejecting failures

“Sort both parts by color”

6 of 8 candidates passed simulation
2 candidates dropped a part

Unreachable task: 8 of 8 rejected
No plan available for approval



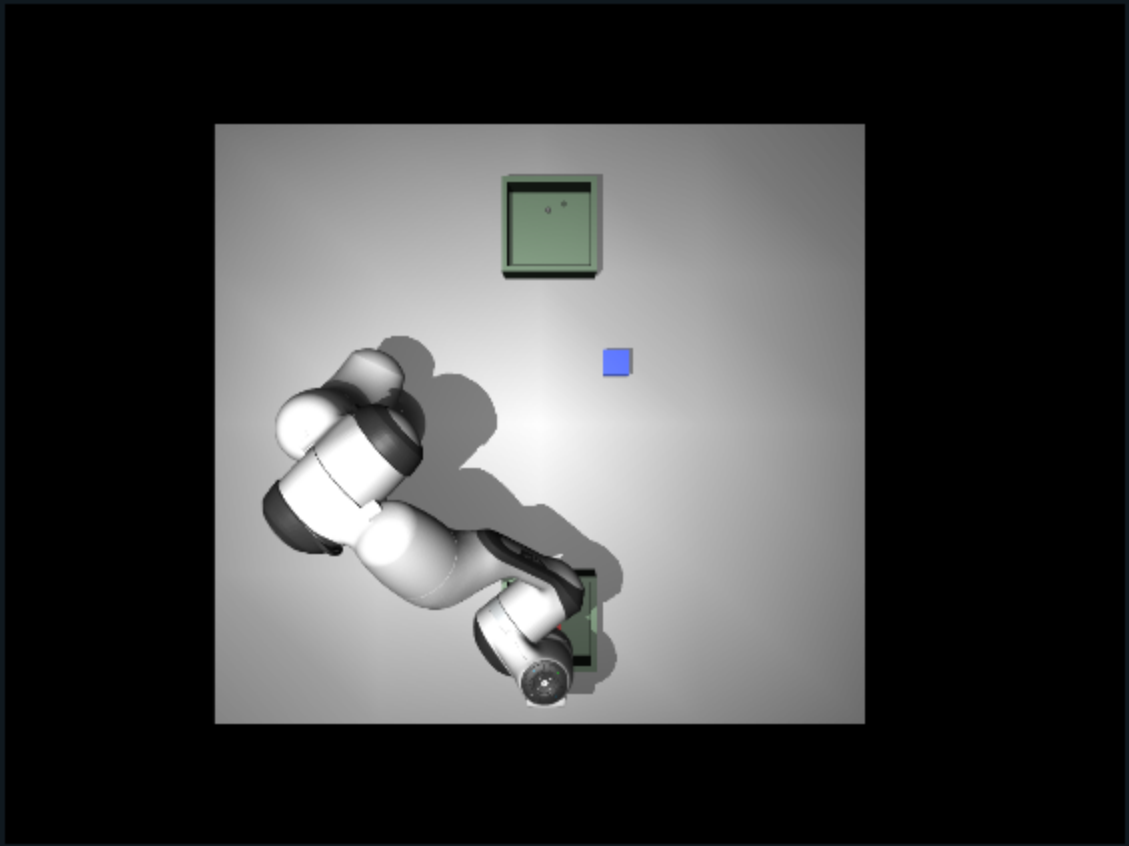
Results from three live AMD runs

Task	Passed / 8	Rejected	Total wall time
Red part to bin A	6	2 dropped parts	20.570 s
Sort both parts	6	2 dropped parts	115.485 s
Blue part out of reach	0	8 unreachable	9.786 s

24 candidates tested. 12 passed simulation. 12 rejected.

Single runs, October 16, 2026. Client wall times include service delays.
Simulation ran locally on macOS. Sorting vision check took 97.555 s.

Camera uncertainty requires human review



Geometry: placement passed

Camera model: “no”

The arm obscures the part.
Both successful live tasks
required human review.

The operator must inspect the evidence and acknowledge review before export.

AMD powers planning and image review

AMD Token Factory

Planner: DeepSeek-V4-Flash
Image review: DeepSeek-V4.1-Flash

Local execution

MuJoCo physics, four simulation workers and a Gradio operator page

The shared API does not report GPU SKU or ROCm version.

Next step: a factory pilot

Test before you touch.

Today: a working simulation and approval prototype

Next: clearer camera views, broader collision checks and calibrated hardware trials

Visionary Creators

github.com/anish-01/simtoreal-agent